

Advanced NLP

Session 13 - Retrieval Augmented Generation (RAG) & Dense Retrieval (DPR)

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Contents

1. Foundations & The Need for Retrieval
2. The Architectural Triad of RAG Design
3. The Retriever Engine: Sparse, Dense & Hybrid
4. LM Integration & Context Pathologies
5. Adaptive RAG, Tool Use & Datastore Scaling
6. The Frontier: Autonomous Deep Research

1.

Foundations & The Need for Retrieval

- From SQuAD Reading Comprehension to Open-Domain QA (DrQA)
- Pathologies of Monolithic LMs: Hallucinations, Cutoffs & Finite Capacity
- The Core RAG Intuition: Parameter Efficiency, Dynamic Updates & Verifiability

Question Answering: Reading Comprehension

Traditional Reading Comprehension

- **Context (SQuAD):** Input is Query Q + explicit reference Passage C.
- **Task:** Locate start and end text spans directly inside passage.
- **Example:** Marco Polo *'through contact with Persian traders'*

Multimodal In-Context QA

- **Input:** Given image (e.g. pizza) + natural prompt.
- **Task:** Perform visual reasoning (count slices, inspect toppings).
- **Key Feature:** Context is provided upfront within the prompt.



Conventional Methods for Reading Comprehension

ERA 1

Feature-Based (Pre-Deep Learning)

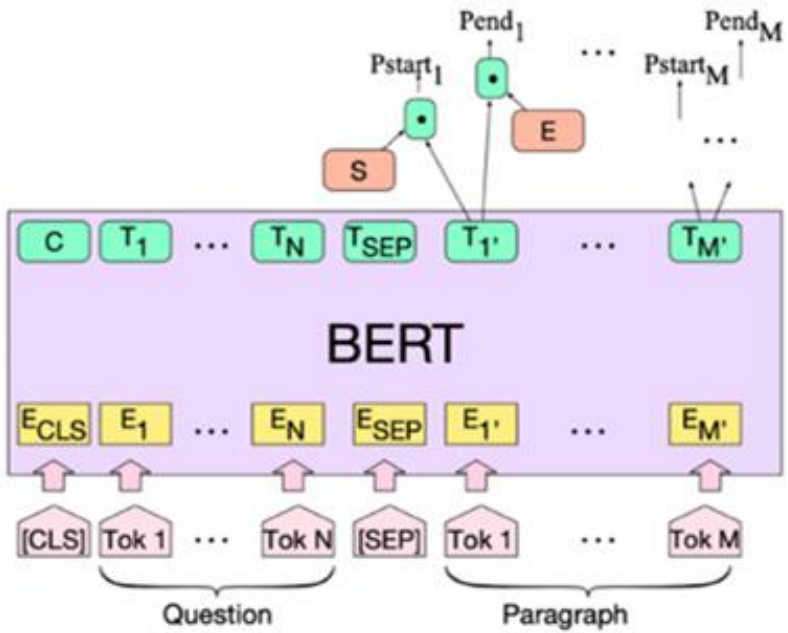
- **Candidate extraction:** (a_1 ... a_M)
- **Feature vectors:** ϕ(p, q, a_i) using n-grams & parse trees
- **Classification:** Multi-class logistic regression

ERA 2

Neural Span Extraction (BERT)

- **Input formatting:** Concatenate Question Q & Context C with [SEP]
- **Span prediction:** Pointer indices (1 ≤ start ≤ end ≤ N)
- **Mechanism:** Joint self-attention over [CLS] Q [SEP] C

Neural Span Extraction Architecture (BERT)



Open-Domain Question Answering (ODQA)

REALISTIC PROBLEM SETTING

- **No Given Context:** No pre-filtered passage provided upfront.
- **Massive Corpus:** Access only to full corpus *D* (e.g., Wikipedia).
- **Unknown Location:** Exact answer location is completely unknown.

OPEN-DOMAIN VS. CLOSED-DOMAIN

- **Closed-Domain:** Restricted to narrow topics (e.g., clinical manuals, tech support).
- **Open-Domain:** Answers arbitrary queries across all human knowledge.
- **Core Challenge:** Finding a needle in a massive information haystack.

Open-Domain QA Pipeline Architecture



Retrieval Augmentation: The DrQA Breakthrough

PIONEERING WORK (CHEN ET AL., 2017)

Reading Wikipedia to Answer Open-Domain Questions

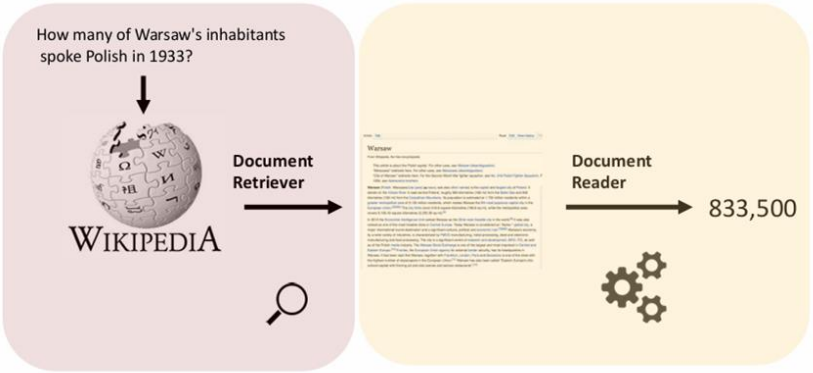
- Query: *'How many of Warsaw's inhabitants spoke Polish in 1933?'*
- Doc Retriever: Scans 5M+ articles → finds 'Warsaw'
- Doc Reader: Neural reader extracts answer → '833,500'

SYSTEM PARADIGM

Decoupled Philosophy

- Memory Offloading: Delegate broad memory to external index
- Focused Neural Compute: Reserve compute for comprehension

DrQA ARCHITECTURE & PIPELINE FLOW



https://github.com/facebookresearch/DrQA

https://github.com/facebookresearch/DrQA

Overcoming LM Shortcomings with Just-In-Time Retrieval

THE FUNDAMENTAL QUESTION

Instead of memorizing everything, can we provide content just-in-time?

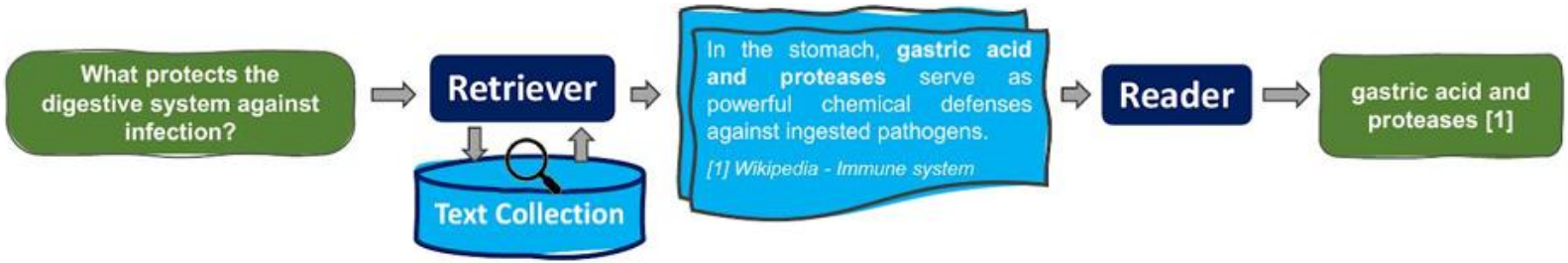
1. DYNAMIC UPDATES

- Easily update or append documents in datastore.
- Zero retraining or weight fine-tuning.

2. INTERPRETABLE VERIFICATION

- Model generates pointers and citations to sources.
- Enables verifiable human auditing.

JUST-IN-TIME RETRIEVAL PIPELINE



Limitations of Monolithic LMs: Confident Hallucinations

EVALUATING GPT-5.1 (TESTED JAN 2026)

Query: *'Who is Akari Asai?'* (CMU NLP Instructor)

Result: Generates an authoritative, highly fluent fake biography.

AUDIT OF GENERATED 'REPRESENTATIVE PAPERS'

Paper 1 (RAG, NeurIPS 2020)

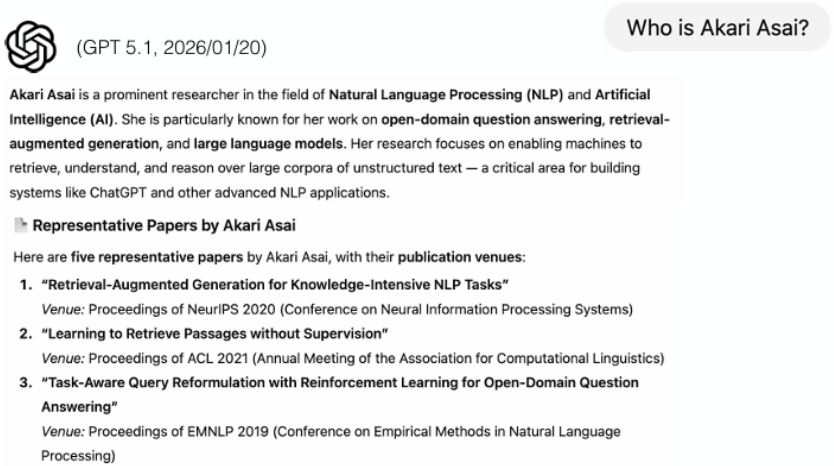
Wrong Attribution — Actually written by Patrick Lewis et al.

Paper 2 (Learning to Retrieve, ACL 2021)

Wrong Attribution — Actually written by Gautier Izacard et al.

Paper 3 (EMNLP 2019)

Completely Fabricated — Does NOT exist at all!



Takeaway: LMs fabricate plausible academic credentials with high confidence.

Hallucinations in Peer-Reviewed Science: ICLR Citation Audit

GPTZero Audit of ICLR Submissions

Papers scoring 8.0 in review contain verified hallucinations!

Case 1: TamperTok (Review Rating: 8.0)

Cited NeurIPS 2023 paper exists, but **all author names are wrong**.

Case 2: MixtureVitae (Review Rating: 8.0)

Cited MMLU paper **hallucinates 7 non-existent co-authors**.

Case 3: Catch-Only-One (Review Rating: 6.0)

Cited paper is **completely fabricated** with zero match.

VERIFIED CITATION AUDIT TABLE (GPTZERO)

ICLR

https://gptzero.me/news/iclr-2026/

Title	Average Review Rating	Paper Link	Citation Check Scan Link	Example of Verified Hallucination	Comment
TamperTok: Forensics-Driven Tokenized Autoregressive Framework for Image Tampering Localization	8.0	TamperTok: Forensics-Driven Tokenized Autoregressive Framework for Image Tampering Localization | OpenReview	https://app.gptzero.me/documents/4645494f-70eb-40bb-aea7-0007e13f7179/share	Chong Zou, Zhipeng Wang, Ziyu Li, Nan Wu, Yuling Cai, Shan Shi, Jiawei Wei, Xia Sun, Jian Wang, and Yizhou Wang. Segment everything everywhere all at once. In Advances in Neural Information Processing Systems (NeurIPS), volume 36, 2023.	This paper exists, but all authors are wrong.
MixtureVitae: Open Web-Scale Pretraining Dataset With High Quality Instruction and Reasoning Data Built from Permissive Text Sources	8.0	MixtureVitae: Open Web-Scale Pretraining Dataset With High Quality Instruction and Reasoning Data Built from Permissive Text Sources | OpenReview	https://app.gptzero.me/documents/bfd10666-ea2d-454c-9ab2-75faa8b84281/share	Dan Hendrycks, Collin Burns, Steven Basart, Andy Critch, Jerry Li, Dawn Ipolito, Aina Lapedriza, Florian Tramer, Rylan Macfarlane, Eric Jiang, et al. Measuring massive multitask language understanding. In Proceedings of the International Conference on Learning Representations (ICLR), 2021.	The paper and first 3 authors match. The last 7 authors are not on the paper, and some of them do not exist
Catch-Only-One: Non-Transferable Examples for Model-Specific Authorization	6.0	Catch-Only-One: Non-Transferable Examples for Model-Specific Authorization | OpenReview	https://app.gptzero.me/documents/9afb1d51-c5c8-48f2-9b75-250d950f6252?share	Dinghuai Zhang, Yang Song, Inderjit Dhillon, and Eric Xing. Defense against adversarial attacks using spectral regularization. In International Conference on Learning Representations (ICLR), 2020.	No Match

Limitations of Monolithic LMs: Obsolete Knowledge

TEMPORAL CUTOFF DILEMMA

Query: *'When did GPT-5.2 come out?'* (Tested Jan 2026)

Result: Model output claims no release date exists (cutoff mid-2024).

GROUND TRUTH REALITY

OpenAI officially released GPT-5.2 on **December 11, 2025**.

RETRAINING DILEMMA

Continual pre-training to update parametric knowledge is compute-prohibitive and causes **catastrophic forgetting**.



Theoretical Root Cause: The Monolithic Pre-Training Loss

AUTOREGRESSIVE PRE-TRAINING LOSS

$$\mathcal{L}(\theta; D) = -\frac{1}{|D|} \sum_{x \in D} \sum_{t=1}^{|x|} \log p_\theta(x_t \mid x_{<t})$$

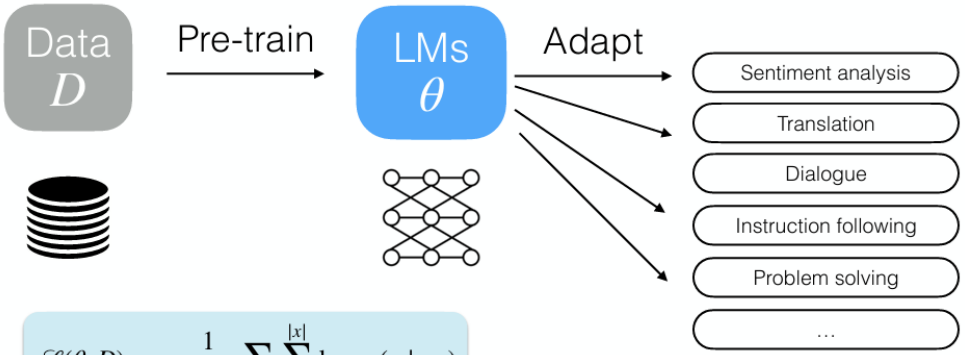
1. PARAMETER CAPACITY LIMIT ()

Cannot store infinite facts in parametric memory. Factual knowledge is strictly bounded by bits-per-parameter scaling laws.

2. CORPUS TEMPORAL LIMIT (D)

Corpus D only captures the state of the world at data cutoff time. New facts after pre-training remain completely inaccessible.

MONOLITHIC PRE-TRAINING & ADAPTATION FLOW



$$\mathcal{L}(\theta; D) = -\frac{1}{|D|} \sum_{x \in D} \sum_{t=1}^{|x|} \log p_\theta(x_t \mid x_{<t})$$

θ can't store infinite information

D only capture information as of the data cutoff time

Core Dilemma: Parametric memory () and training corpus (D) create a static, closed system that fails on dynamic knowledge updates.

RAG Intuition 1: Eliminating Hallucinations & Parameter Efficiency

Without Retrieval (Parametric Only)

Query: *'Who is Akari Asai?'*

Output: 'Japanese voice actress' ✗

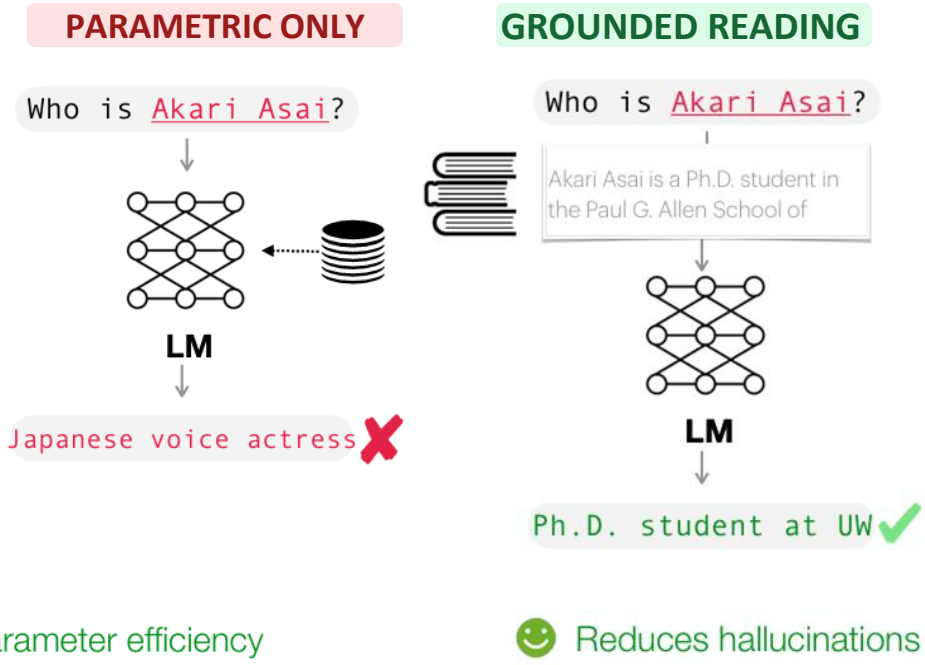
With Retrieval (Grounded Reading)

Retrieved text: *'Ph.D. student in Allen School...'*

LM output: 'Ph.D. student at UW' ✓

Core Gains

- Drastically reduces factual hallucinations
- Parameter efficiency: Compact LMs with retrieval beat massive models



RAG Intuition 2: Real-Time Updates & Verifiability

WITHOUT RETRIEVAL (STATIC PARAMETRIC)

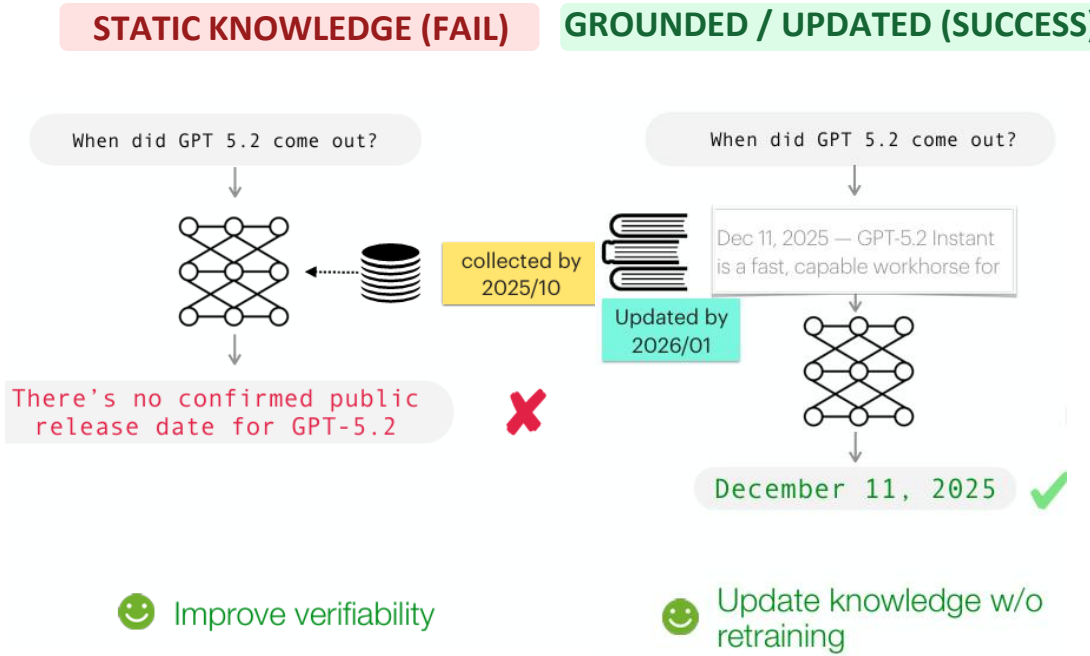
- Frozen data up to 2025/10 → Fails on release date [✗]

WITH RETRIEVAL (DYNAMIC KNOWLEDGE)

- Datastore updated in 2026/01 with latest web release notes
- LM reads new doc → 'December 11, 2025' [✓]

CORE GAINS

- Update world knowledge dynamically without retraining weights
- Improve verifiability with auditable source citations



Formalizing the Retriever-Reader Framework

1. Inputs & Outputs

Corpus D

D = {D1, ..., DN}

Query Q

User input question

Answer String A

Target output response

2. Formulations

Retriever

f(D, Q) → {P1, ..., PK}

(Top candidate passages, K ~ 100)

Reader

g(Q, {P1, ..., PK}) → A

(Reading comprehension task)

3. Implementations

Retriever Options

- Fixed sparse BM25
- Learned dense dual encoder

Reader Options

- Extractive BERT
- Generative autoregressive LLM

Retriever: $f(\mathcal{D}, Q) \longrightarrow P_1, \ldots, P_K$

Reader: $g(Q, \{P_1, \ldots, P_K\}) \longrightarrow A$

K is pre-defined (e.g., 100)

A reading comprehension problem!

The Seminal RAG Architecture (Lewis et al., 2020)

1. Non-Parametric Memory

Dense Document Index queried via MIPS

2. Parametric Memory

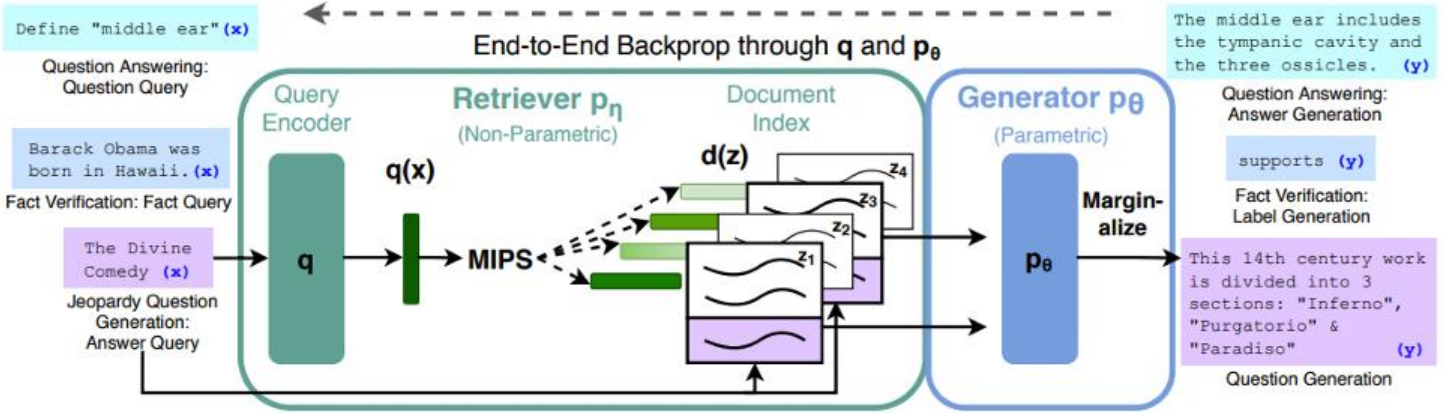
Pre-trained seq2seq Generator *p*

3. Latent Variable Marginalization

pRAG(y|x) i z top-k p(z|x) p(yi|x, z, y1:i-1)

4. Joint Optimization

End-to-end backprop through query encoder *q(x)* & generator *p*



Step-by-Step Scoring Mechanics in Modern RAG

Input Query **'When did GPT 5.2 come out?'**

STEP 1 **Compute Similarity Scores** Sim(d | x)

Doc 1 (OpenAI release notes) **0.9 • Relevant**

Doc 2 (Ruby on Rails 5.2) **0.1 • Distractor**

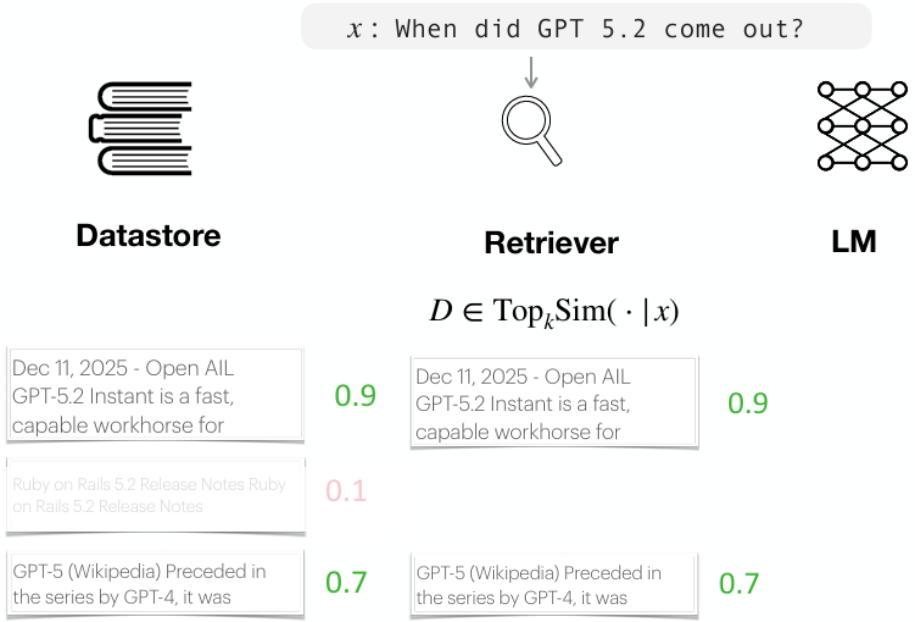
Doc 3 (Wikipedia GPT-5) **0.7 • Close**

Top-k Filtering

Drops **Doc 2**; forwards **Doc 1 & 3** to Reader.

Synthesis

LM generates final answer: **'Dec, 2026'**



2.

The Architectural Triad of RAG Design

- Pillar 1: Datastore Engineering (Sources, Chunking, Scaling)
- Pillar 2: Retrieval Architecture (Sparse, Dense, Rerankers)
- Pillar 3: Language Model Integration (Prompting, Layer Fusion, Output Blending)

The Triad of RAG Design Choices: Part 1 - Datastore

1. Datastore Sources

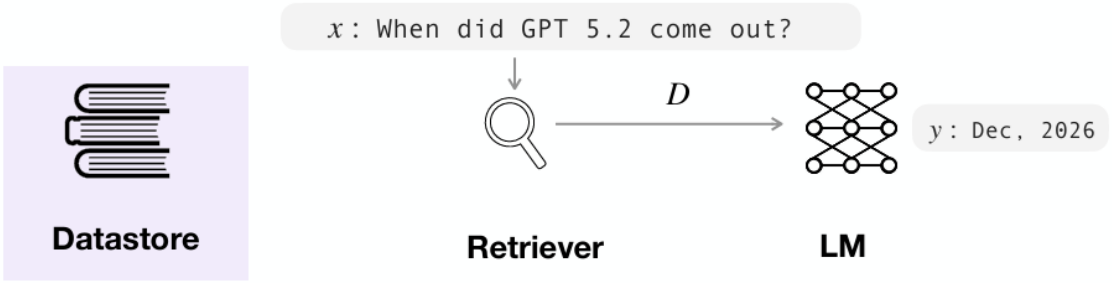
Web crawl, Wikipedia, GitHub code, documentation, forums.

2. Processing & Chunking

Delimiter-based vs. fixed token windows.

3. Scaling Laws

Expanding datastores from billions to trillions of tokens.



- ✓ Sources of datastore
- ✓ Processing
- ✓ Scaling

The Triad of RAG Design Choices: Part 2 - Retriever

1. Types of Retrievers

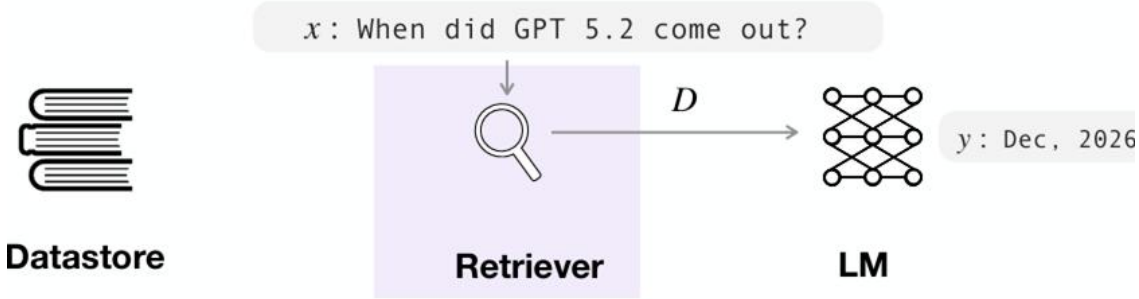
Sparse (BM25) vs. Dense (DPR) vs. Rerankers.

2. Training Methodologies

Dual-encoder contrastive InfoNCE loss.

3. Evaluation Metrics

nDCG@10, BEIR, MTEB.



- ✓ Types of retrievers
- ✓ Training
- ✓ Evaluations

The Triad of RAG Design Choices: Part 3 - Language Model

1. Integration Modes

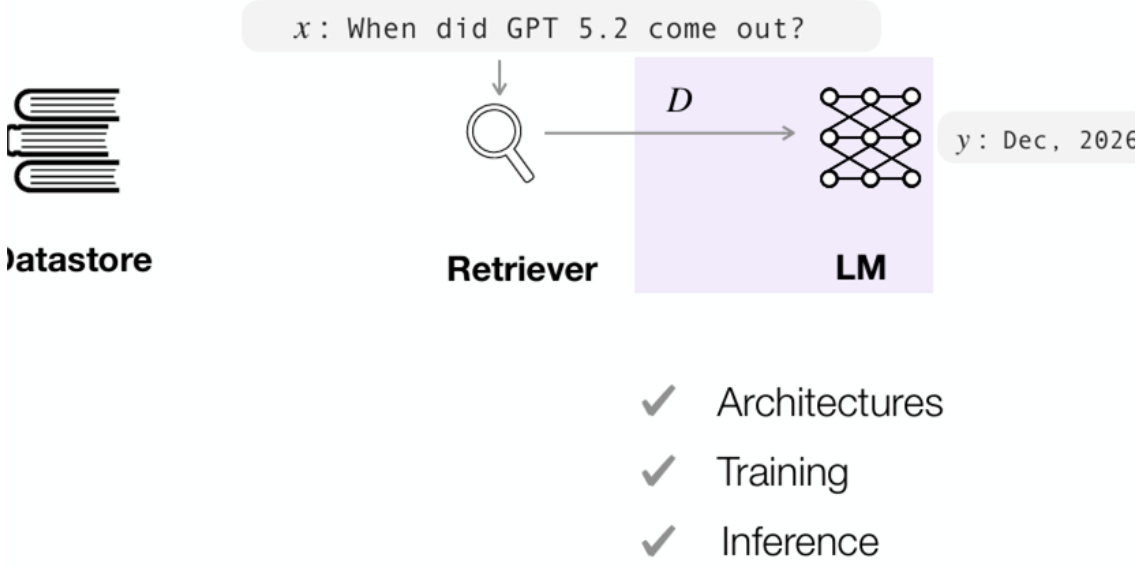
Prompt augmentation vs. intermediate cross-attention vs. output blending.

2. Training Paradigms

Independent vs. sequential vs. joint end-to-end.

3. Inference Adaptation

Self-RAG reflection tokens and autonomous research agents.



3.

The Retriever Engine: Sparse, Dense & Hybrid

- Lexical Representations: Bag-of-Words, TF-IDF & BM25 Mathematical Framework
- The Lexical Pitfall: Why Word Overlap Fails at Semantic Meaning
- Dense Vector Retrieval: Encoders, InfoNCE Contrastive Learning & Cross-Encoders

Core Question: How Should We Retrieve Passages from Large Corpora?

Retriever Architectures: The Three Core Families

01 Lexical Systems

e.g., BM25

Word Overlap

Key Mechanism:

Exact keyword matching performed via inverted indexes.

Primary Advantage:

⚡ Ultra-fast search speed with zero neural inference overhead.

02 Dense Vector

DPR, Contriever

Single Dense Vector

Key Mechanism:

Neural dual encoders mapping queries and passages into a shared dense vector space.

Primary Advantage:

🧠 Deep semantic matching beyond exact lexical overlaps.

03 Multi-Vector

e.g., ColBERT

Late Interaction

Key Mechanism:

Token-level contextualized representations combined with fine-grained late interaction operators.

Primary Advantage:

🎯 High expressivity of cross-encoders with efficient retrieval.

SEARCH & RETRIEVAL PARADIGMS

Interaction Paradigms: Trade-offs in Speed vs. Expressivity

(a) Bi-Encoder

- Independent vectors, fast dot product
- **Misses** token interactions

(b) Interaction Matrix

- Match matrices between Q & D
- CNN / MLP scoring layers

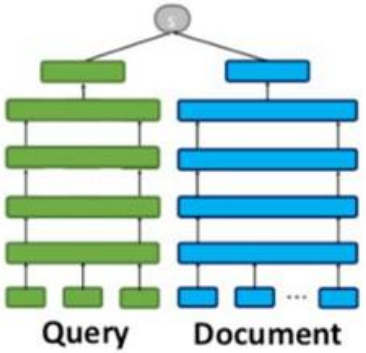
(c) Cross-Encoder

- All-to-all token attention
- High quality, **slow** for large corpus

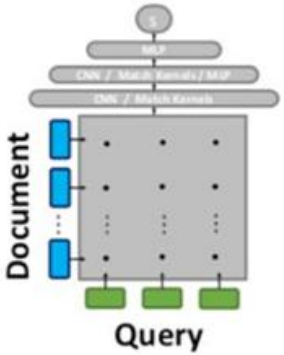
(d) Late Interaction

- ColBERT: MaxSim Operator

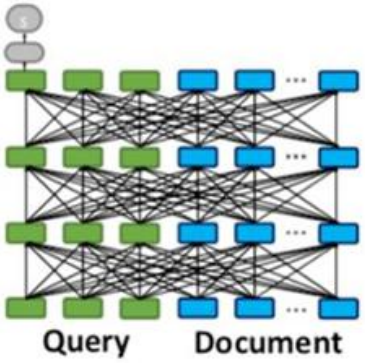
$$Score(Q,D) = \sum max(E_Qi \cdot E_Dj^T)$$



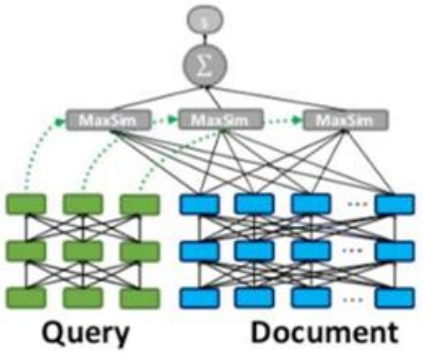
(a) Representation-based Similarity
(e.g., DSSM, SNRM)



(b) Query-Document Interaction
(e.g., DRMM, KNRM, Conv-KNRM)



(c) All-to-all Interaction
(e.g., BERT)



(d) Late Interaction
(i.e., the proposed ColBERT)

WEAKLY SUPERVISED ODQA

Learned Retrieval: Joint Training in ORQA

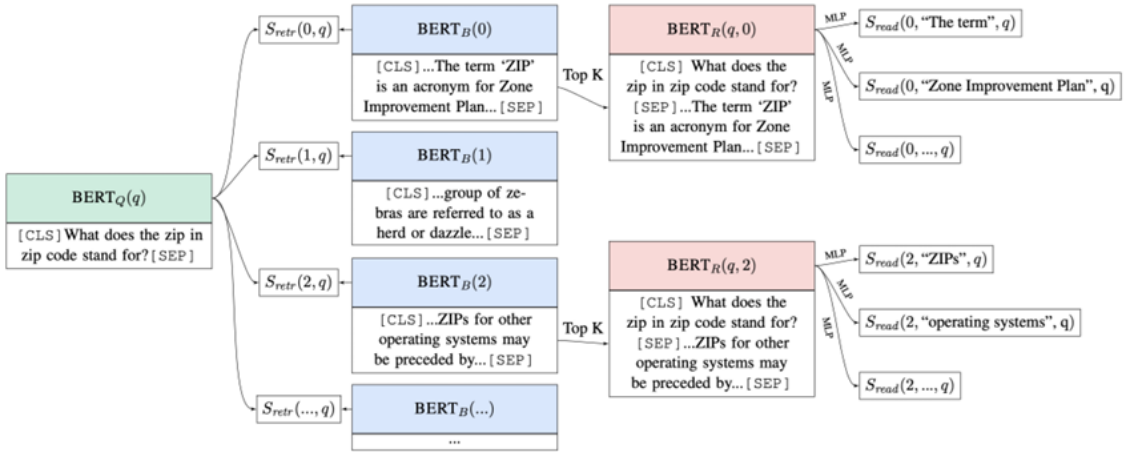
Latent Retrieval (Lee et al., 2019)

Jointly trains retriever and reader without requiring explicit passage labels.

Architecture Mechanics

- **BERT_Q & BERT_B** compute S_retr → Top K passages
- Concatenate with query → **BERT_R** predicts answer span S_read

ORQA PIPELINE



⚠️ Computational Bottleneck

Updating passage encoder requires rebuilding index over **21M documents**.

DENSE PASSAGE RETRIEVAL (DPR)

Empirical Milestone: 1k Q/A Pairs Beat BM25

Supervised Dual-Encoder

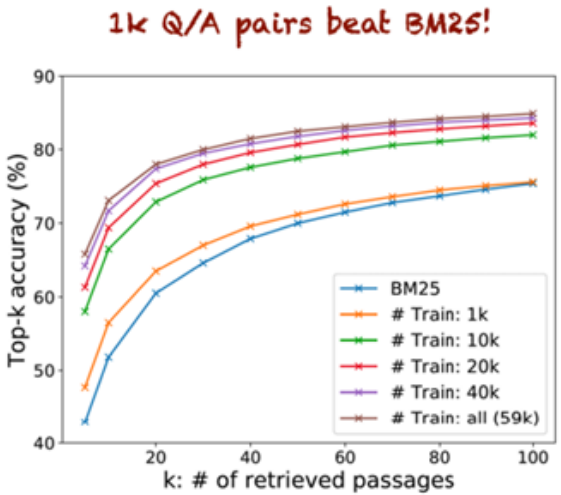
Karpukhin et al., 2020

- **Similarity:** sim(q, p) = h_q^T h_p
- **Encoders:** BERT_Q (question) & BERT_P (passage)

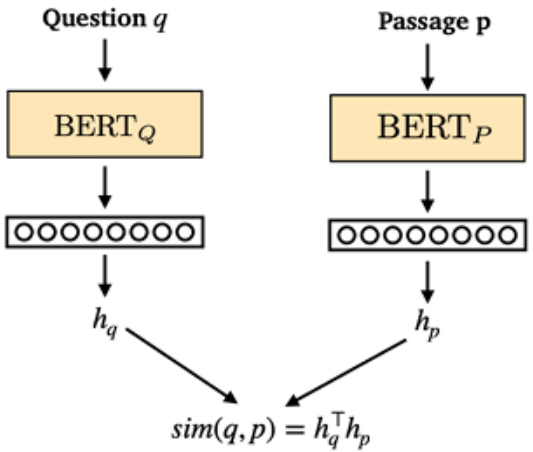
Empirical Milestone

- **1,000 Q/A pairs** (orange curve) beat baseline BM25 (blue curve)!
- Scaling to 59k pairs achieves **>85% top-100** retrieval accuracy.

Retrieval Accuracy vs. Training Data



Dual-Encoder Architecture



Retriever Tiers & Compute Budget

Cost vs. Performance Hierarchy of Retrievers

Sparse Retrievers $

Lookup: Inverted index (TF-IDF, BM25)

Extremely fast & CPU-friendly matching.

Dense Retrievers $$

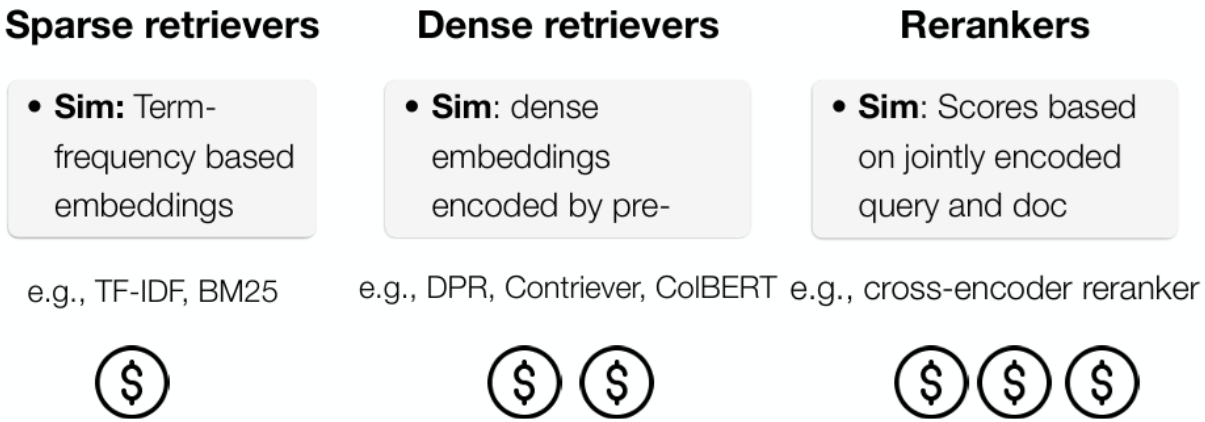
Lookup: Dual-encoders (DPR, Contriever)

Vector index search via FAISS.

Rerankers $$$

Lookup: Cross-encoders

Joint attention over (query, document) pairs.



SPARSE REPRESENTATIONS

Binary Term Matching

01 Toy Corpus Setup

Query q: 'what is nlp'

d_1: 'what is life ? candy is life !'

d_2: 'nlp is an acronym for natural language'

d_3: 'I like to do good research on'

02 Binary Incidence Check

- Presence Check: Term presence encoded as binary {0, 1} vector.
- Limitations: Ignores term frequency, document length, and word importance.

TERM-DOCUMENT INCIDENCE MATRIX

	q=what is nlp	d_1 = what is life ? candy is life !	d_2 = nlp is an acronym for natural language	d_3 = I like to do good research on
what	1	1	0	0
candy	0	1	0	0
nlp	1	0	1	1
is	1	1	1	0
language	0	0	0	0
life	0	1	0	0
…	…	…	…	…

Sparse Representations

One-Hot Lexical Vectors

01 Orthogonal Vector Space

Axis Assignment

Each vocabulary term is assigned an independent, mutually orthogonal axis.

Extreme Dimensionality

Vector dimension equals vocabulary size: **$|V| \geq 50{,}000+$** dimensions.

```
v("nlp") = [ 1, 0, 0, 0, ... 0 ]
v("candy") = [ 0, 1, 0, 0, ... 0 ]
v("lang") = [ 0, 0, 1, 0, ... 0 ]
```

02 Severe Semantic Limitation

Equidistant Term Distances

Distance between related terms **'nlp'** & **'natural language'** is identical to completely unrelated terms **'nlp'** & **'candy'**.

Zero Semantic Topology

No concept of similarity, context, or semantic overlap between distinct lexical tokens.

Dot Product = 0 for any non-identical pair

(Complete structural isolation of words)

SPARSE REPRESENTATIONS

Term-Count Vectors

Frequency-Based Counting

Vector entries equal raw occurrence count in document *d*.

d_1: 'what is life ? candy is life !' → **count('life') = 2**

d_2: 'nlp is an acronym for natural language' → 'nlp'=1, 'is'=1

TERM-DOCUMENT COUNT MATRIX

	q=what is nlp	d_1 = what is life ? candy is life !	d_2 = nlp is an acronym for natural language	d_3 = I like to do good research on
what	1	1	0	0
candy	0	1	0	0
nlp	1	0	1	1
is	1	1	1	0
language	0	0	0	0
life	0	2	0	0
…	…	…	…	…

Count the number of appearances in a doc

Major Limitation

Stopwords dominate: Frequent words like *'is'* overwhelm vectors, while longer documents accumulate unfairly high counts.

SPARSE REPRESENTATIONS

Mathematical Formulations: TF-IDF & BM25

01 TF-IDF Model

- **Normalized Term Frequency (TF):**

$$\mathrm{TF}(t,d) = \frac{\mathrm{freq}(t,d)}{\sum_{t'} \mathrm{freq}(t',d)}$$

- **Inverse Document Frequency (IDF):**

$$\mathrm{IDF}(t) = \log\left(\frac{|D|}{\sum_{d' \in D} \delta(\mathrm{freq}(t,d') > 0)}\right)$$

d_1 = what is life ?
candy is life !

$$\mathrm{TF}(t,d) = \frac{\mathrm{freq}(t,d)}{\sum_{t'} \mathrm{freq}(t',d)} \qquad \mathrm{IDF}(t) = \log\left(\frac{|D|}{\sum_{d' \in D} \delta(\mathrm{freq}(t,d') > 0)}\right)$$

t_1 = what

of documents

of documents **where term t appears**

$$\text{TF-IDF}(t,d) = \mathrm{TF}(t,d) \times \mathrm{IDF}(t)$$

02 BM25 Probabilistic Framework

- **Robertson et al. (2009) Ranking Function:**

$$\text{BM-25}(t,d) = \mathrm{IDF}(t) \cdot \frac{\mathrm{freq}(t,d) \cdot (k_1 + 1)}{\mathrm{freq}(t,d) + k_1 \cdot (1 - b + b \cdot \frac{|d|}{\mathrm{avgdl}})}$$

Incorporates non-linear TF saturation (k_1) and document length normalization (b).

$$\text{BM-25}(t,d) = \mathrm{IDF}(t) \cdot \frac{\mathrm{freq}(t,d) \cdot (k_1 + 1)}{\mathrm{freq}(t,d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{\mathrm{avgdl}}\right)}$$

SPARSE REPRESENTATIONS

Weighted-Term Vectors: Step 1 - Stopword Variation

Query Computation

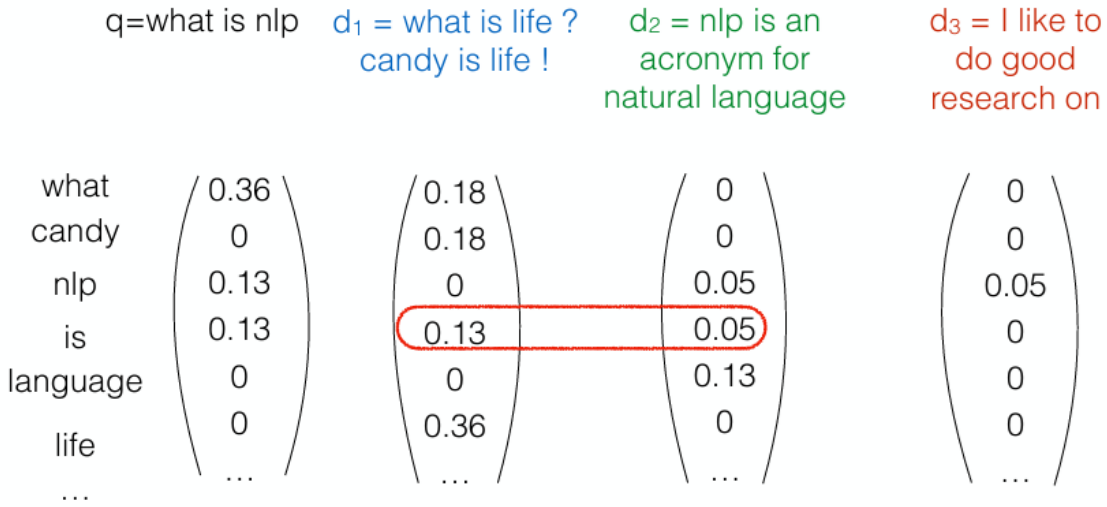
Query **q** ('what is nlp'):
'what' = 0.36 | 'nlp' = 0.13 | 'is' = 0.13

Observation on Term 'is'

Short Doc d_1: 'is' weight = **0.13**
Longer Doc d_2: 'is' weight drops to **0.05**

Key Takeaway

TF-IDF successfully penalizes frequent words in long documents.



Compute TF-IDF weights to build weighted vectors

SPARSE REPRESENTATIONS

Weighted-Term Vectors: Step 2 - Target Passage Sparsity

Analyzing Document d_2 (True Def)

Text: *'nlp is an acronym for natural language'*

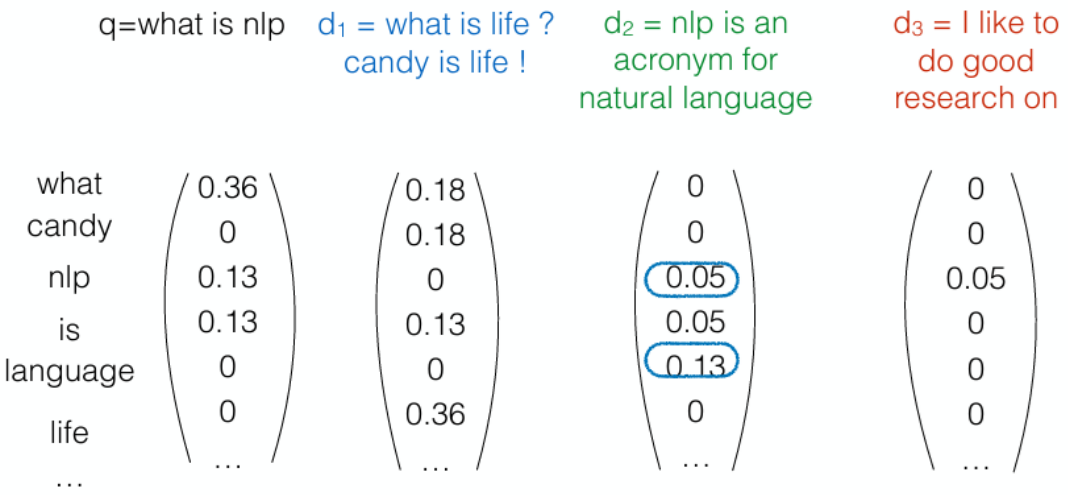
Term Weights: 'nlp'=0.05 | 'is'=0.05 | 'language'=0.13

The Missing Dimension

Query word **'what'** has weight = **0** in d_2 because 'what' is absent literally.

Key Diagnostic Verdict

Demonstrates **lexical vulnerability** in pure sparse term weighting.



Compute TF-IDF weights to build weighted vectors

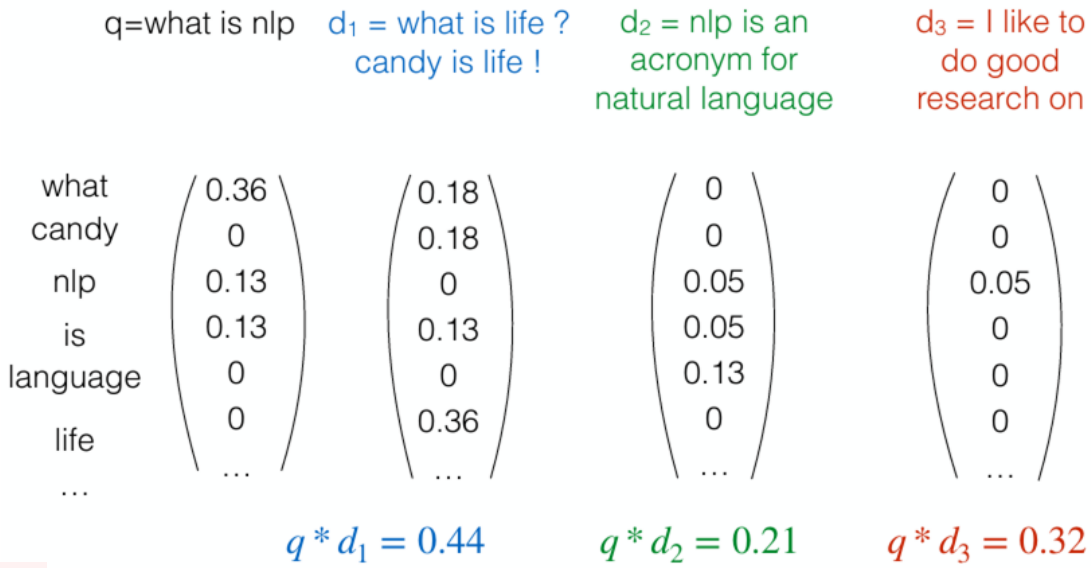
Cosine Similarity Scoring: The Distractor Triumphs

Computing Dot Products (q · d)

d_1 (Rank 1 - Winner!): $q \cdot d_1 = \mathbf{0.44}$
'what is life? candy is life!'

d_3 (Rank 2): $q \cdot d_3 = \mathbf{0.32}$

d_2 (Rank 3 - Lowest!): $q \cdot d_2 = \mathbf{0.21}$
'nlp is an acronym for natural language'



The Failure Mode (Anomaly)

The irrelevant 'candy' distractor document beats the true definition by **more than 2x**!

LEXICAL RETRIEVAL LIMITATIONS

The Lexical Pitfall: Semantic Blindness

Core Diagnostic Verdict

Lexical matching cannot capture semantic similarities or topical intent across query and document spaces.

Strategic Conclusion

Sparse IR is inherently blind to meaning — transition to dense vector representations is mandatory.

The Synonymy & Paraphrase Problem

Identical semantic concepts produce zero sparse lexical match:

Term Pair A

'Natural Language Processing' vs **'Computational Linguistics'**

Term Pair B

'US President' vs **'Commander in Chief'**

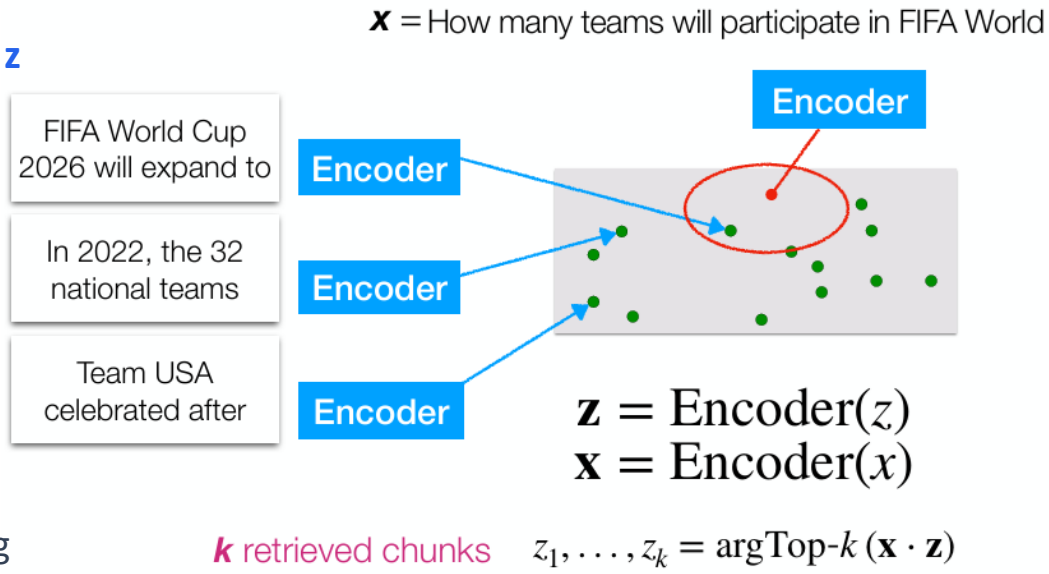
Dense Retrievers: Mapping Text into Latent Vector Spaces

Continuous Semantic Geometry

- Text chunk **z** → dense embedding **z = Encoder(z)**
- Query **x** → dense embedding **x = Encoder(x)**

Vector Retrieval via Inner Product

- Retrieve Top-k chunks via MIPS: **z_1 ... z_k = argTop-k(x · z)**
- Semantically related texts cluster together regardless of exact wording

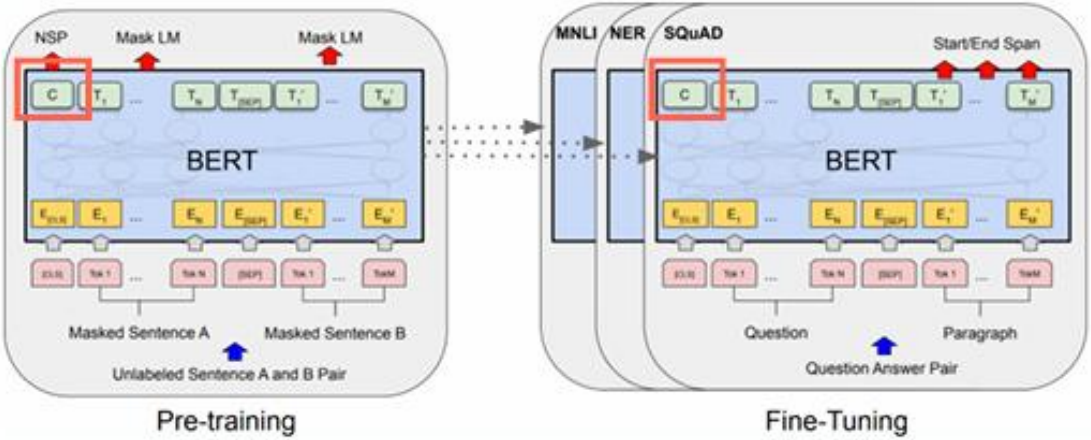


EMBEDDING EXTRACTION STRATEGIES

Dense Embedding Extraction: [CLS] vs. Pooling

Strategy 1: [CLS] Token Representation (R^d)

- Final-layer output of [CLS] token in bidirectional Masked LMs
- Standard in BERT and DPR



Strategy 2: Mean / Max Pooling ($R^{Nxd} \rightarrow R^d$)

- Average or max across all token states
- Applicable to causal autoregressive LMs (SBERT, SGPT, Qwen-Embedding)

CONTRASTIVE LEARNING OBJECTIVE

Training Dense Retrievers: InfoNCE Loss

Dual-Encoder Optimization

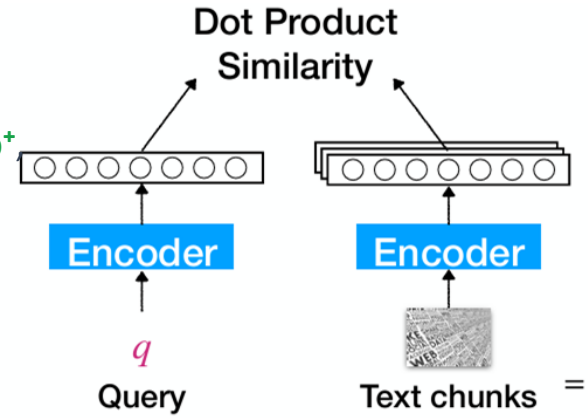
- Input tuple: Query **q**, Positive passage **p^+**, and Negatives **$\{p_1^-, \ldots, p_n^-\}$**

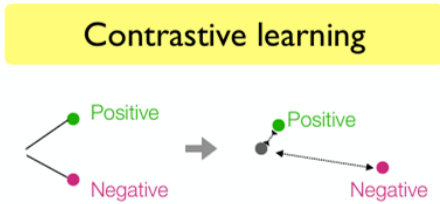
Multi-Class Contrastive Loss

- Maximizes log-likelihood of positive passage over negative distractors

Vector Space Dynamics

- **Attracts:** Pulls query and positive passage together
- **Repels:** Pushes negative distractors far apart





$$L(q, p^+, p_1^-, p_2^-, \ldots, p_n^-) = -\log \frac{\exp(\text{sim}(q, p^+))}{\exp(\text{sim}(q, p^+)) + \sum_{i=1}^{n} \exp(\text{sim}(q, p_j^-))}$$

$$\mathcal{L}(q, p^+, p_1^-, \ldots, p_n^-) = -\log \frac{\exp(\text{sim}(q, p^+))}{\exp(\text{sim}(q, p^+)) + \sum_{j=1}^{n} \exp(\text{sim}(q, p_j^-))}$$

GENERATIVE PASSAGE FUSION

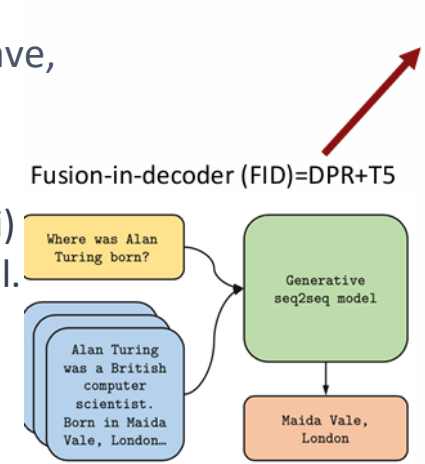
Fusion-in-Decoder (FiD) Architecture

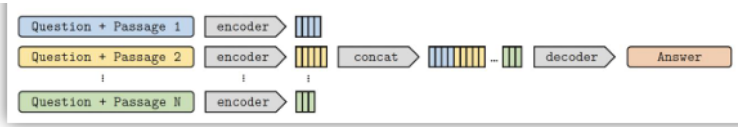
From Extraction to Generation

Combines DPR retrieval with T5 seq2seq generation (Izacard & Grave, 2020).

Decoupled Linear Complexity

- Encode (Question + Passage_i) pairs independently in parallel.
- Concatenate representations; fuse only in Decoder cross-attention.





Model	NaturalQuestions	TriviaQA	
ORQA (Lee et al., 2019)	31.3	45.1	-
REALM (Guu et al., 2020)	38.2	-	-
DPR (Karpukhin et al., 2020)	41.5	57.9	-
SpanSeqGen (Min et al., 2020)	42.5	-	-
RAG (Lewis et al., 2020)	44.5	56.1	68.0
T5 (Roberts et al., 2020)	36.6	-	60.5
GPT-3 few shot (Brown et al., 2020)	29.9	-	71.2
Fusion-in-Decoder (base)	48.2	65.0	77.1
Fusion-in-Decoder (large)	**51.4**	**67.6**	**80.1**

State-of-the-Art Performance

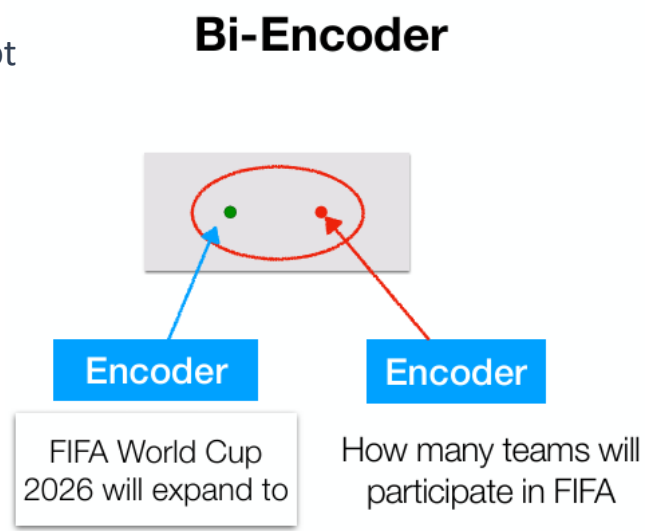
- NaturalQuestions: **51.4%** (vs 44.5% RAG)
- TriviaQA: **80.1%**

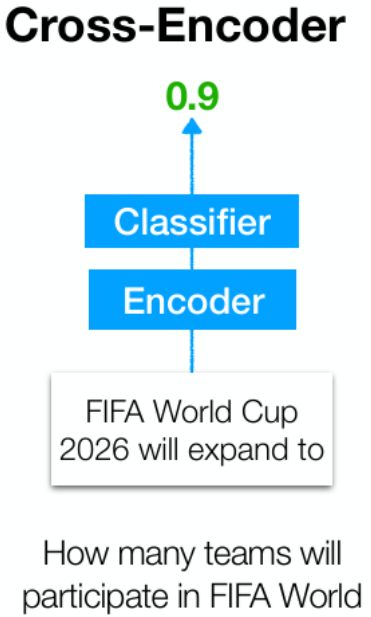
TWO-STAGE RETRIEVAL & RERANKING

Precision Reranking with Cross-Encoders

Bi-Encoder vs. Cross-Encoder

- **Bi-Encoder:** Separate encoders, fast dot product, misses token-to-token attention.
- **Cross-Encoder:** Concatenates query & document into a single Transformer.
  - Full cross-attention across all tokens.
  - Deep relevance classification score (e.g., **0.9**).





Industry Standard Pipeline

Bi-Encoder retrieves **Top-100**
→ Cross-Encoder reranks to **Top-5**

4.

LM Integration & Context Pathologies

- Integration Taxonomy: Input Augmentation vs. Intermediate Fusion (RETRO)
- The Long-Context Trap: 'Lost in the Middle' & Diminishing Returns of Depth
- The Grounding Paradox: Citation Hallucinations & Commercial Search Engine Audits

Input Augmentation Bottlenecks: Scalability & Grounding

Standard In-Context RAG Diagnosis

Prepends retrieved passages **D** directly to the Language Model input prompt.

VULNERABILITY 01

Not Scalable to Many Documents

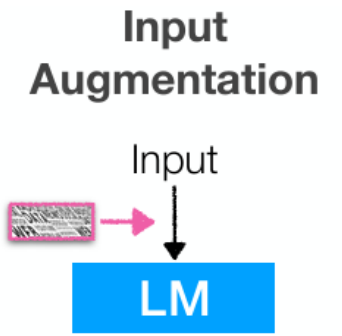
- **Quadratic Attention Costs:** $O(N^2)$ computational overhead.
- **Strict Context Window Limits:** Hard prompt boundaries.
- **Prompt Overhead:** High latency & API costs.

VULNERABILITY 02

Not Strictly Grounded

- **Context Ignoring:** Model may ignore provided context.
- **Hallucination Risk:** Generates plausible but false info.
- **Attention Drift:** Degrades with larger prompt sizes.

Input Augmentation Diagram



- Augment input of LMs
- Easy to apply (w/o training) & effective
- Difficulty of using many D

e.g., RAG

How to Use Retrieval: Input Augmentation

CONCEPT & MECHANISM

In-Context Prompt Augmentation (Standard RAG)

Prepend retrieved text chunks **D** directly to LM input prompt.

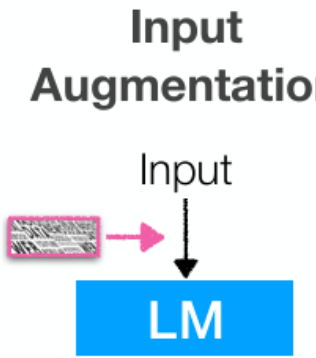
ADVANTAGES

- **Zero training required** on base LM: works with any off-the-shelf model.
- **Highly effective** for general QA.

KEY TRADE-OFF

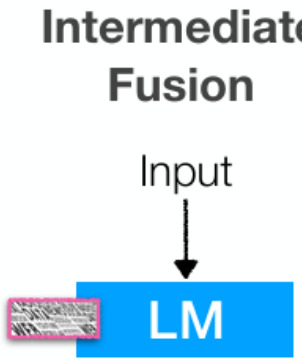
Bounded by context window size

Suffers from attention degradation as context length grows.



- Augment input of LMs
- Easy to apply (w/o training) & effective
- Difficulty of using many D

e.g., RAG



- Modify LMs to incorporate D in intermediate layers
- Scalable to many passages
- Requires retraining

e.g., RETRO, InstructRETRO

How to Use Retrieval: Intermediate Fusion

CONCEPT & MECHANISM

Architectural Layer Fusion

Modify internal Transformer layers to incorporate retrieved passages via cross-attention inside hidden layers.

ADVANTAGES

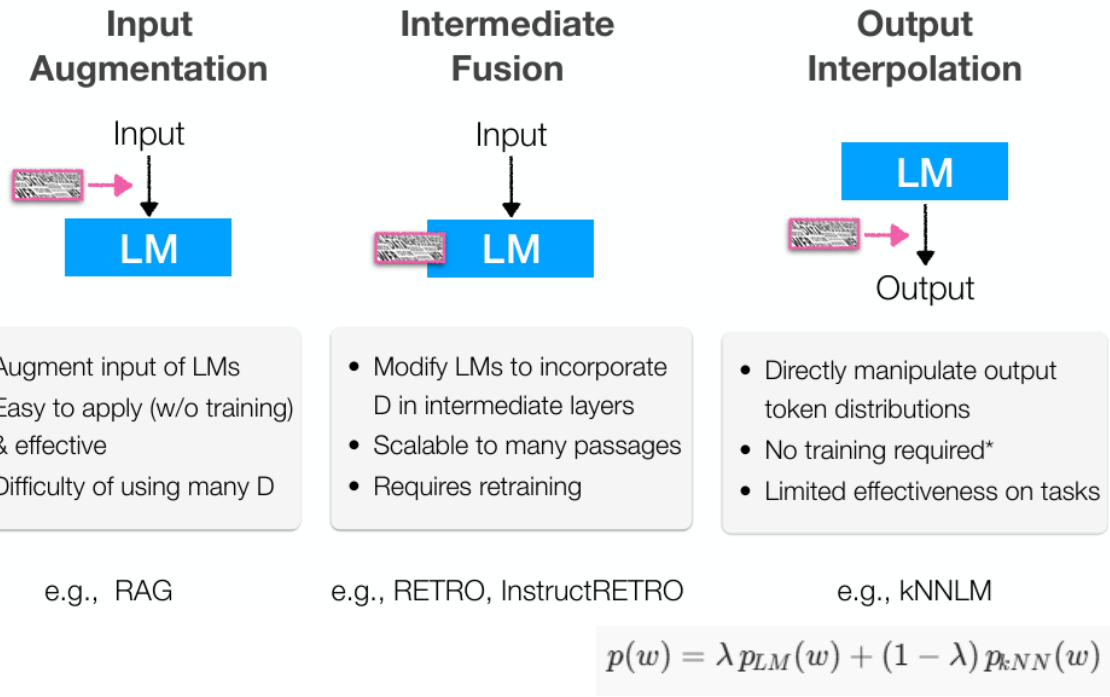
High Scalability

Scalable to hundreds of passages without prompt bloat.

KEY TRADE-OFF

Custom Architecture Required

Requires pre-training or fine-tuning model architecture (e.g., RETRO).



DeepMind RETRO: Trillion-Token Retrieval via CCA

RETRIEVAL AT SCALE

Trillion-Token Datastore (Borgeaud et al., 2022)

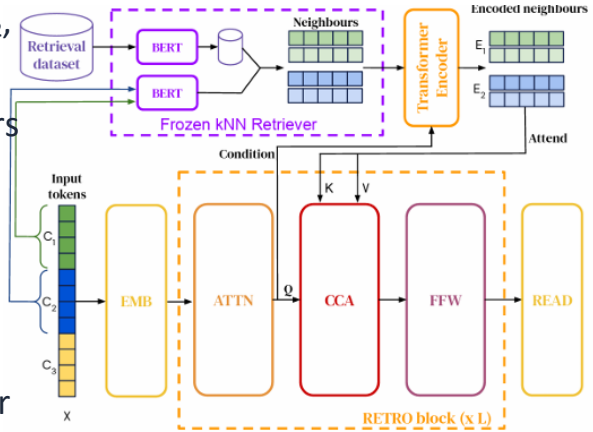
- Input split into **64-token chunks** (C_1, C_2, C_3)
- **Frozen kNN retriever** fetches neighbors from multi-trillion token datastore

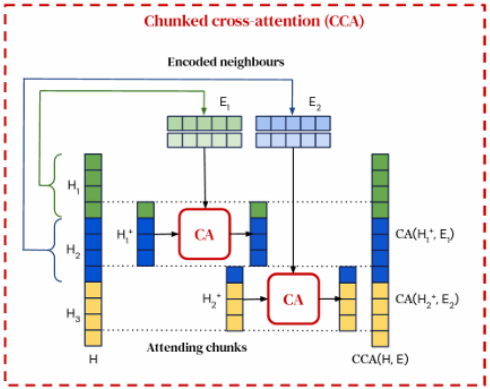
CHUNKED CROSS-ATTENTION (CCA)

Architectural Layer Fusion

- Intermediate layers attend to neighbor embeddings (E_1, E_2) via **CCA blocks**
- **Causality preserved:** tokens attend only to neighbors of preceding chunks

ARCHITECTURE & CCA PIPELINE





Left: Chunk retrieval & neighbor encoding via Transformer

Right: CCA blocks integrating neighbor embeddings into LM

The Context Capacity Question: Why Not Stuff More Docs?

1. NAIVE ENGINEERING INTUITION

More Passages = Better Coverage?

"If 5 passages give 70% recall, retrieve 30 passages to guarantee full coverage."

2. THE PROMPT SETUP

Stuffing 30 Search Results

Provide documents 1–30 in prompt and instruct model: *"Answer using only provided search results."*

3. THE LURKING TRAP

Attention Allocation Failure

How do LLMs actually allocate attention across dozens of stuffed context documents?

PROMPT STUFFING STRUCTURE (NOBEL PRIZE EXAMPLE)

Input Context

```
Write a high-quality answer for the given question using only the provided search
results (some of which might be irrelevant).

Document [1](Title: Asian Americans in science and technology) Prize in physics for
discovery of the subatomic particle J/ψ. Subrahmanyan Chandrasekhar shared...
Document [2](Title: List of Nobel laureates in Physics) The first Nobel Prize in
Physics was awarded in 1901 to Wilhelm Conrad Röntgen, of Germany, who received...
Document [3](Title: Scientist) and pursued through a unique method, was essentially
in place. Ramón y Cajal won the Nobel Prize in 1906 for his remarkable...

Question: who got the first nobel prize in physics
Answer:
```

Desired Answer

```
Wilhelm Conrad Röntgen
```

Positional Bias: The 'Lost in the Middle' Phenomenon

1. EXPERIMENTAL AUDIT

Long-Context Attention Setup (Liu et al., 2023)

Exactly 1 gold passage hidden among 10, 20, or 30 distractor documents.

2. THE U-SHAPED ACCURACY CURVE

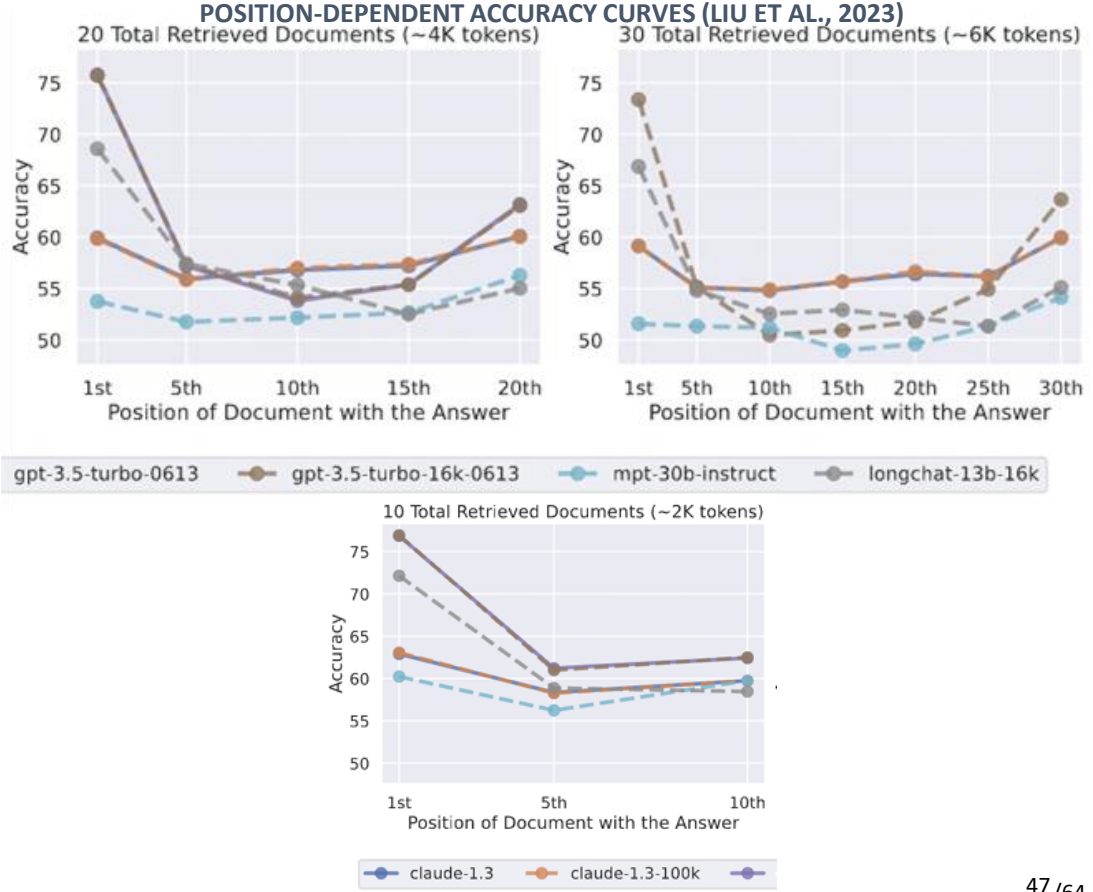
Primacy Bias: High accuracy when answer is at the beginning.

Recency Bias: Moderate accuracy when answer is at the end.

Middle Plunge: Severe accuracy drop (~50%) when buried in the middle!

3. CORE TAKEAWAY

Large context windows ≠ effective context utilization.



Empirical Saturation: The Diminishing Returns of Passage Depth

1. RECALL VS. ACCURACY GAP

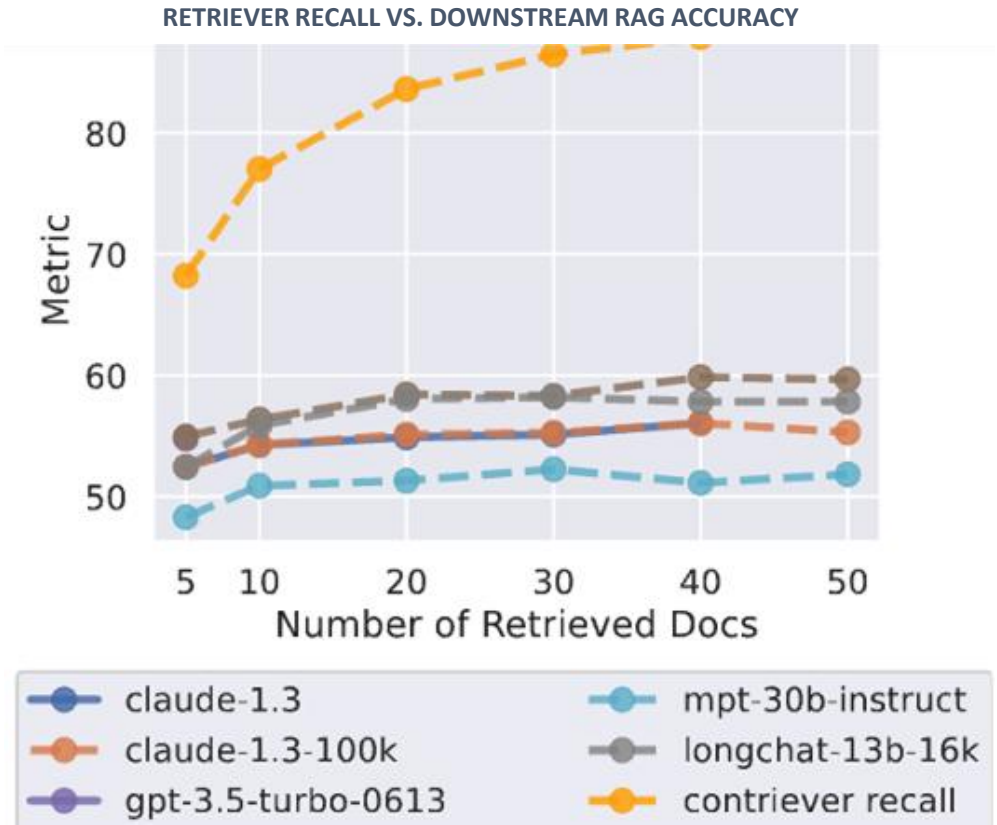
Recall Climbs, Accuracy Plateaus

- **Retriever Recall (Yellow)**: Climbs towards 90% at 50 documents.
- **Downstream RAG Accuracy (Solid)**: Saturates and plateaus after 10–20 documents!

2. THE NOISE BARRIER

Distractor Noise Overwhelms Evidence

- Adding passages increases distractor noise faster than useful evidence.
- Demands **precision filtering** over brute-force context stuffing.



Web-Scale RAG in Commercial Search Engines

1. CONSUMER LANDSCAPE

Systems that search and synthesize the live web with inline citations.

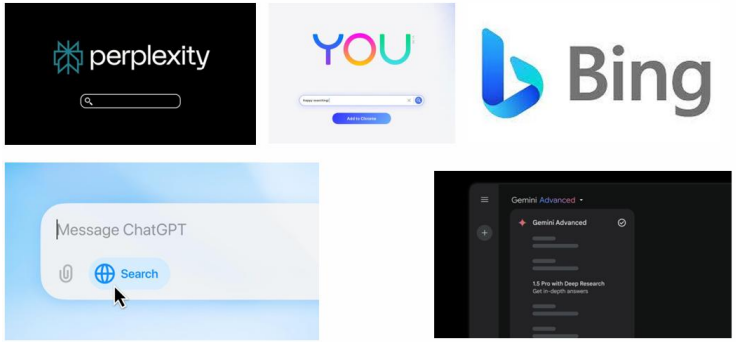
2. PIONEERING PLATFORMS

Perplexity AI, You.com, Copilot (Bing), ChatGPT Search, Gemini Advanced.

3. THE CORE QUESTION

The Promise: Real-time web intelligence backed by source links.

The Question: Can we actually trust these generated citations?



Post-Hoc Verification: RARR (Research and Revise)

RARR FRAMEWORK (GUO ET AL., 2023)

Step 1: Query Gen

Extract claims from LM output → generate targeted search queries.

Step 2: Retrieval

Fetch relevant web snippets for each extracted factual claim.

Step 3: Agreement

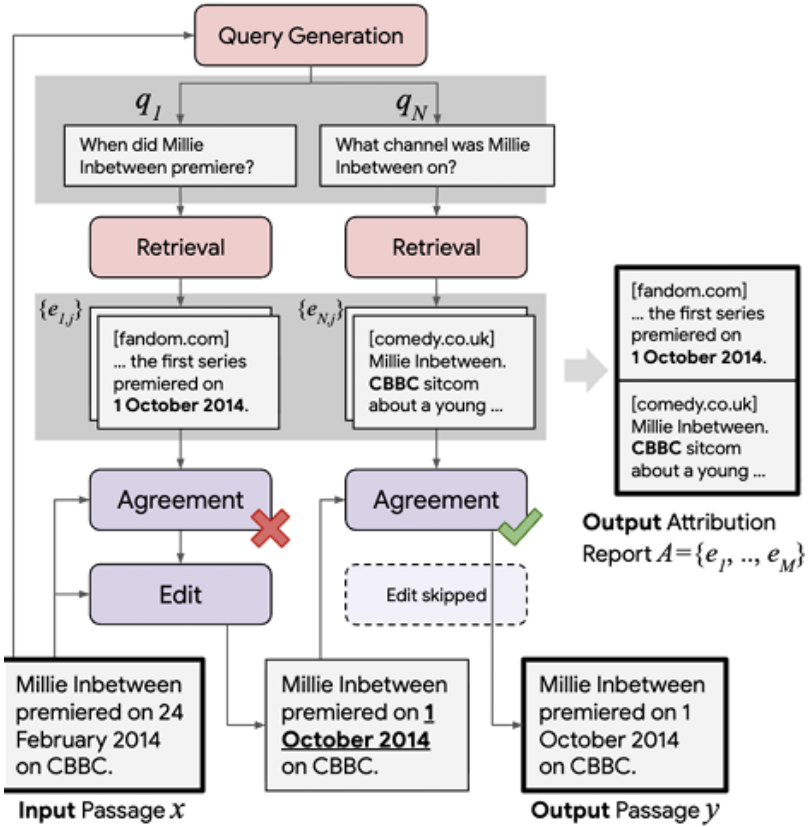
Detect semantic disagreement between generated text and retrieved sources.

Step 4: Edit

Automatically revise refuted text while preserving original voice.

PRIMARY OUTPUT

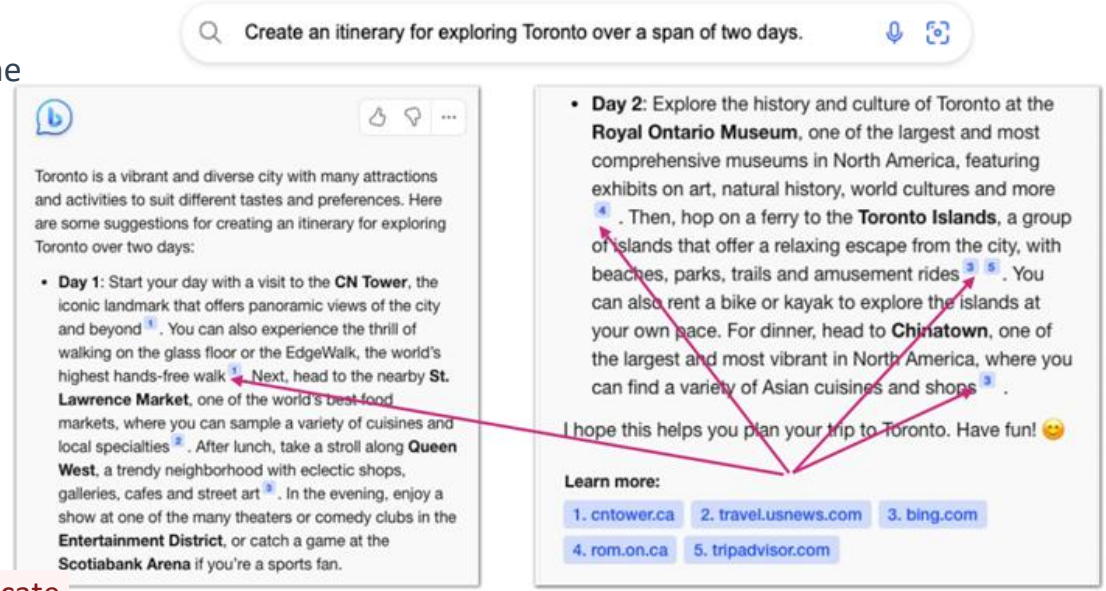
Generates a fully verified **'Attribution Report'** alongside revised text.



The Grounding Paradox: Citation Hallucinations

THE ARCHITECTURAL VULNERABILITY

- Core RAG Promise: Citing verified sources with numbered footnotes.
- Pipeline Reality: Citations are generated by the LLM itself during synthesis.
- **Consequently: Citations themselves can be completely hallucinated!**



THE CRITICAL QUESTION

How often do commercial web-search LLMs fabricate citations?

Citation Audit: Fluent Prose vs. Low Evidentiary Recall

AUDIT: BING, NEEVA, PERPLEXITY, YOUCHAT

Human Perception Metric (High)

Perceived Utility: 4.50 / 5.0
Fluency: 4.48 / 5.0

Objective Evidentiary Metric (Low)

- Citation Precision: 74.5%
(1 in 4 citations unsupported)
- Citation Recall: 51.5%
(Nearly half generated claims unsupported!)

VERDICT

Surface fluency masks severe factual unreliability.

Outputs are easy to read / appear useful to rater (1-5 scale)

Perceived Utility (↑)	Average Over All Queries
Bing Chat	4.34
NeevaAI	4.48
perplexity.ai	4.56
YouChat	4.62
Average	4.50

Fluency (↑)	Average Over All Queries
Bing Chat	4.40
NeevaAI	4.43
perplexity.ai	4.51
YouChat	4.59
Average	4.48

But precision and recall are both low..

Citation Precision (%; ↑)	Average Over All Queries
Bing Chat	89.5
NeevaAI	72.0
perplexity.ai	72.7
YouChat	63.6
Average	74.5

Citation Recall (%; ↑)	Average Over All Queries
Bing Chat	58.7
NeevaAI	67.6
perplexity.ai	68.7
YouChat	11.1
Average	51.5

When to Trust Retrieval: The Long-Tail Knowledge Regime

MEMORY & POPULARITY SPECTRUM

Mallen & Asai et al. (2023)

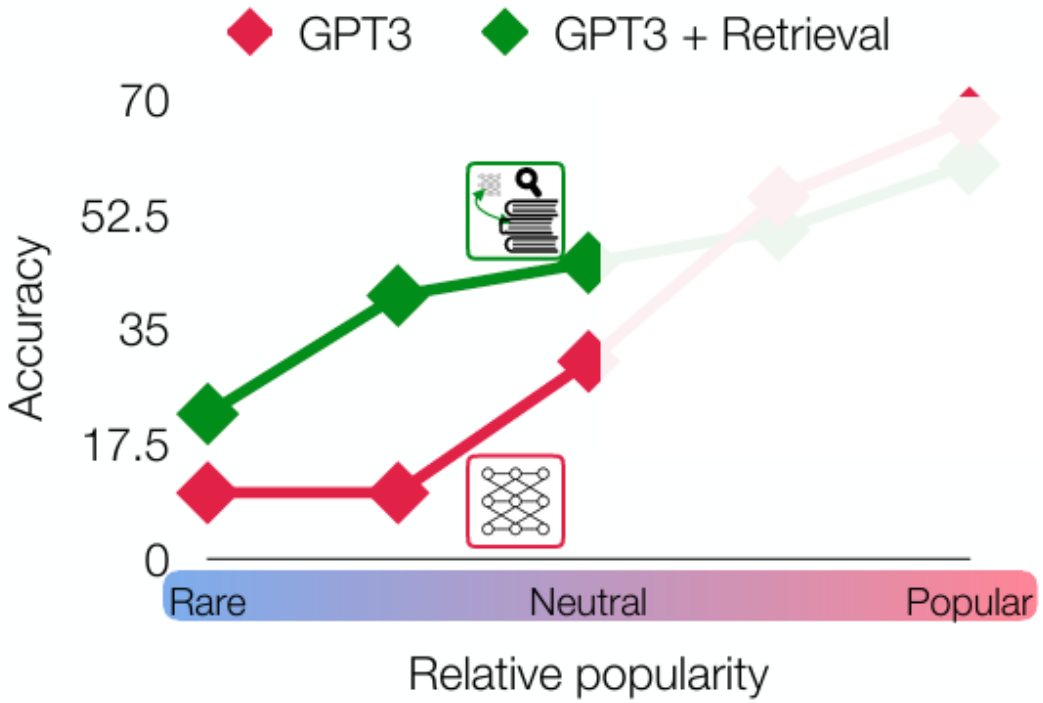
Spectrum: Rare (Long-Tail) → Neutral → Popular

THE LONG-TAIL VICTORY

- **Standalone GPT-3:** Near-zero accuracy for rare long-tail entities.
- **+ Retrieval (RAG):** Delivers massive accuracy improvements!

KEY TAKEAWAY

RAG is essential for specialized, rare, and long-tail domain knowledge.



When NOT to Retrieve: Corruption of Known Entities

POPULAR ENTITY FAILURE MODE

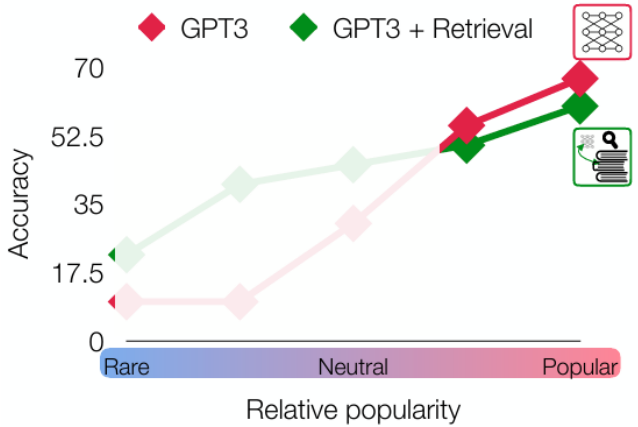
- Standalone GPT-3: Achieves ~70% accuracy on popular entities.
- Unconditional Retrieval: Forces extra context and *degrades* performance!

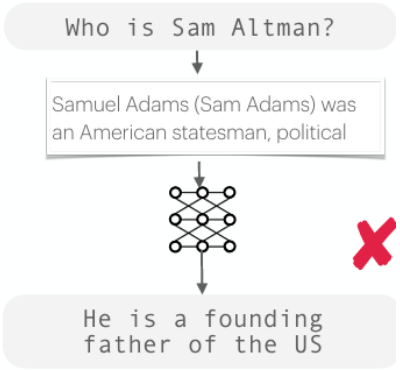
THE SAM ALTMAN VS. SAMUEL ADAMS HAZARD

- Model already knows Sam Altman.
- Unconditional search fetches bio of *'Samuel Adams'*.
- Model gets confused: *"He is a founding father of the US"* ❌

VERDICT

Models must decide WHEN to retrieve.





5.

Adaptive RAG, Tool Use & Datastore Scaling

- Self-RAG: Learning to Retrieve, Generate & Critique via Reflection Tokens
- Toolformer: Self-Supervised Interleaving of Diverse External API Calls
- Non-Parametric Scaling Laws: MassiveDS 1.4-Trillion Token Corpus

Self-RAG: Adaptive Retrieval via Reflection Tokens

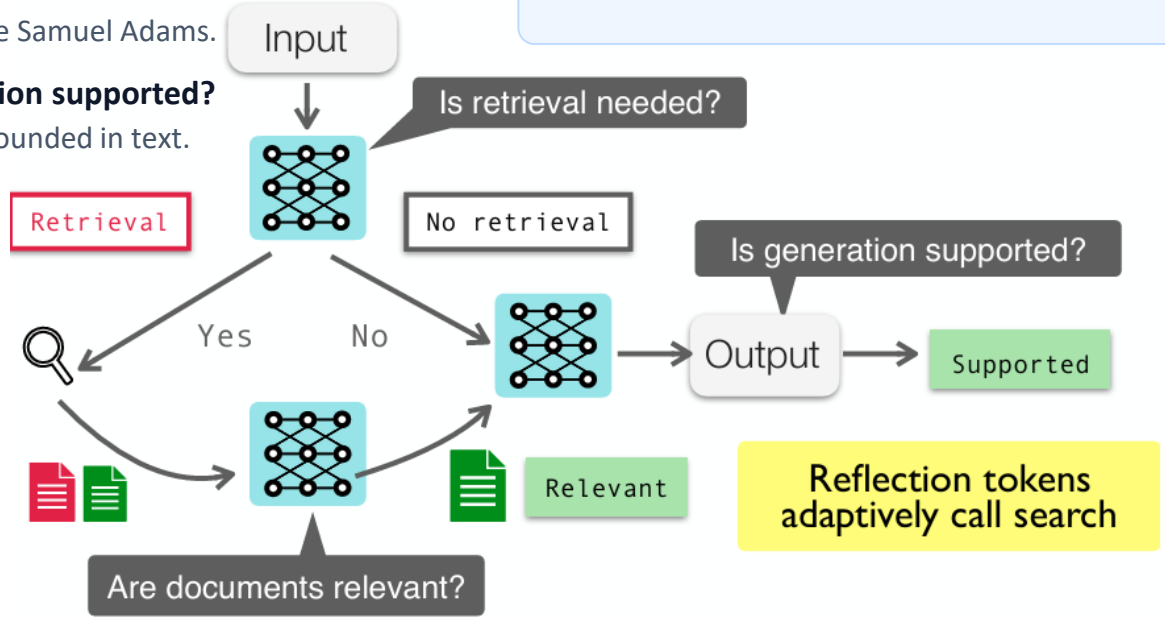
3 ADAPTIVE REFLECTION GATES

Gate 1: Is retrieval needed?
If No, generate directly; if Yes, invoke search.

Gate 2: Are documents relevant?
Filter distractors like Samuel Adams.

Gate 3: Is generation supported?
Verify claims are grounded in text.

Reflection Tokens
Special tokens trained to dynamically evaluate decisions and guide adaptive retrieval at inference time.



Tool-Augmented LMs: Generalizing Retrieval to Diverse Tools

01

Self-Taught Tool Use

Schick et al., 2023

Retrieval is generalized as merely one external API tool among many available APIs.

02

Toolformer Paradigm

Self-Supervised Learning

LMs sample API calls inline, execute them, and filter to keep only calls that demonstrably reduce loss.

03

Diverse Ecosystem

Multi-Tool Integration

WikiSearch for facts, Calculator for arithmetic, and Machine Translation (MT) for translation.

The Brown Act is California's law [WikiSearch("Brown Act") → The Ralph M. Brown Act is an act of the California State Legislature that guarantees the public's right to attend and participate in meetings of local legislative bodies.] that requires legislative bodies, like city councils, to hold their meetings open to the public.

Out of 1400 participants, 400 (or [Calculator(400 / 1400) → 0.29] 29%) passed the test.

The name derives from "la tortuga", the Spanish word for [MT("tortuga") → turtle] turtle.

Matching Corpus to Downstream Task

CORE PRINCIPLE
RAG quality is strictly bounded by datastore coverage.

- **General World Knowledge**
  English Wikipedia dumps (Chen 2017; Lewis 202
- **Technical Code Synthesis**
  GitHub code repositories & debugging
- **Framework Implementation**
  LangChain & LlamaIndex documentation
- **Developer Troubleshooting**
  Stack Overflow community forums

x : when did GPT 5.2 come out?

English Wikipedia



Chen et al., 2017; Gu et al., 2020; Asai et al., 2020; Guu et al., 2021; Lewis et al., 2021 … etc

https://dumps.wikimedia.org/

x : How should I implement RAG using LlamaIndex?

Code snippets



Documentations

LangChain



Community forums

stackoverflow

Scaling Datastores: MassiveDS & Trillion-Token Laws

CORE PARADIGM SHIFT

Non-parametric datastore scaling provides a direct, effective alternative to model parameter bloat.

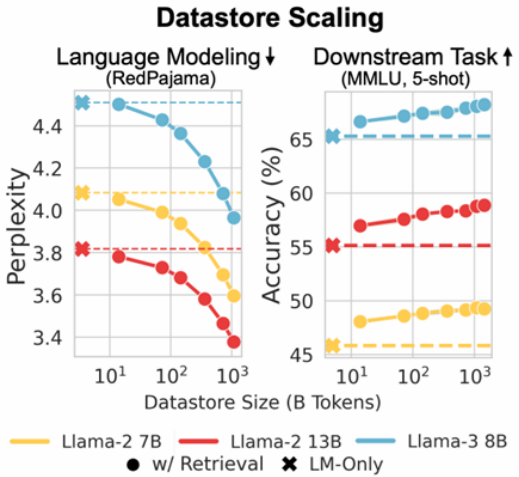
Wikipedia → MassiveDS (Shao et al., 2024)

Scaling datastores from 5 Billion tokens (Wikipedia) to 1.4 Trillion tokens across web, patents, and research papers.



Trillion-Token Scaling Laws

- **Perplexity:** Drops sharply as datastore size expands.
- **Task Accuracy:** Steady MMLU gains across Llama-2 & Llama-3.



6.

The Frontier: Autonomous Deep Research

- From Single-Turn RAG to Multi-Hour Autonomous Inquiry
- Training Deep Research Agents: Large-Scale SFT + Verifiable RL (RLVR)
- Evaluation: Humanity's Last Exam

Frontiers: Autonomous Deep Research (DR) Agents

PARADIGM SHIFT

From Single-Turn Retrieval to Multi-Hour Autonomous Research

Key Systems: OpenAI Deep Research, Gemini Deep Research, DR-Tulu (Shao & Asai, 2025)

Execution Capabilities in Action

Target Query: `"What are the differences between AI2's OLMo2 and OLMo3?"`

STEP 01

Multi-Step Search Tree

Plans dynamic exploration paths and multi-hop web queries.

STEP 02

Document Ingestion

Reads and parses dozens of dense technical reports.

STEP 03

Conflict Resolution

Cross-references evidence and resolves contradictions across sources.

STEP 04

Authoritative Synthesis

Drafts a comprehensive, well-cited synthesis report.

Training Deep Research Agents: SFT + Verifiable RL (RLVR)

STAGE 01

Large-Scale SFT on Teacher Trajectories

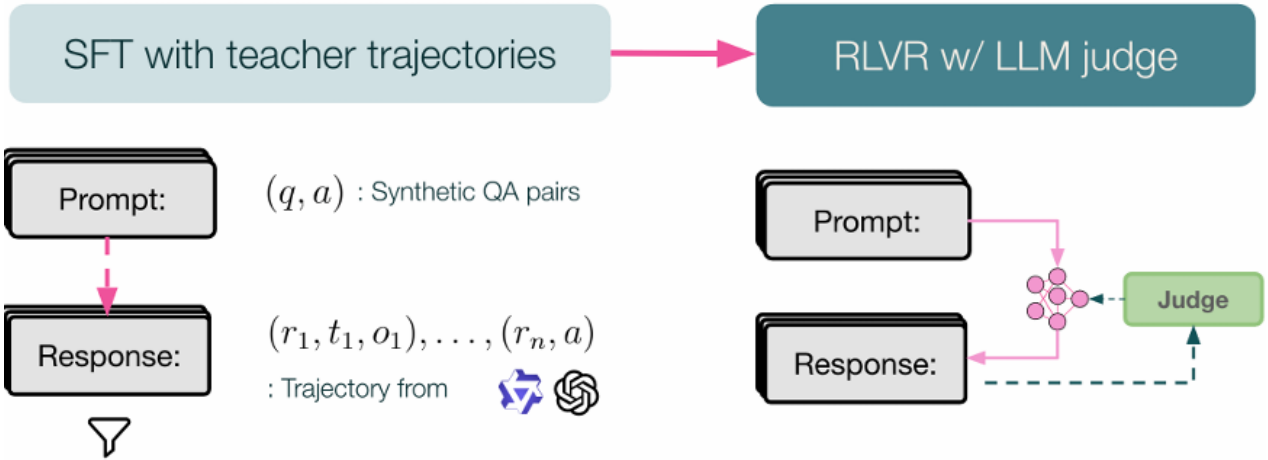
Distill reasoning (ri), tool calls (ti), and observations (oi) via rejection sampling.

STAGE 02

RLVR (Reinforcement Learning w/ Verifiable Rewards)

Agent explores autonomously; rewarded when final output matches verifiable facts.

KEY CAPABILITY: Enables models to master long-horizon information seeking.



Evaluating Deep Research: Humanity's Last Exam & BrowseComp

Deep Research Bench

Evaluating comprehensive, long-form synthesis capabilities across autonomous agents.

Humanity's Last Exam

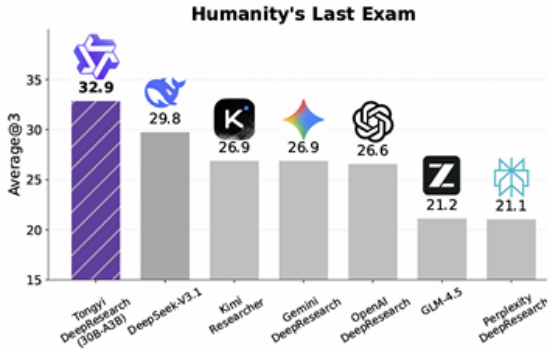
Graduate-level multidisciplinary reasoning benchmark.

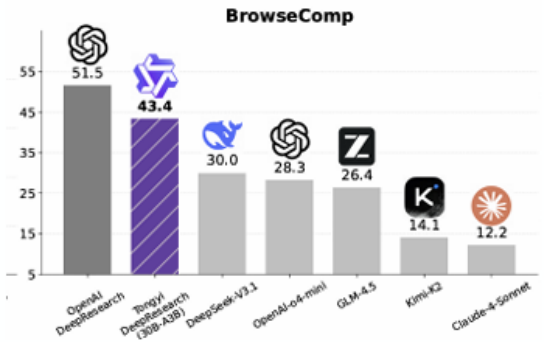
Top: Tongyi DeepResearch-30B-A3B (32.9) beats DeepSeek-V3.1 (29.8) & Gemini (26.9)

BrowseComp

Complex multi-hop web browsing and verification.

Frontier Leaders: OpenAI DeepResearch (51.5) & Tongyi (43.4)





Advanced NLP - Session 13 - Retrieval Augmented Generation (RAG) & Dense Retrieval (DPR) - IUST

Foundations | Architectural | Retriever | Context Pathologies | Adaptive RAG | Deep Research